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## Technical note

## Filtering of electromyogram artifacts from the electrocardiogram

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**Abstract**

Electromyogram (EMG) artifacts often contaminate the electrocardiogram (ECG). They are more difficult to suppress or eliminate, compared for example to the power line interference, due to their random character and to the considerable overlapping of the frequency spectra of ECG and EMG signals obtained from the same pair of electrodes. The usually applied low-pass filtering (cutoff frequency of minimum 35 Hz) results in limited suppression of the EMG artifact and considerable reduction of sharp Q, R and S ECG wave amplitudes. A solution to this problem is proposed by applying approximation filtering with dynamically varied number of samples and weighting coefficients, depending on the ECG signal slope. The slope measure used is the absolute value of the product of the tilts of two adjacent 10 ms segments sliding along the signal. The results obtained show a slight widening of some sharper QRS complexes, but a virtual preservation of their amplitudes and a considerable reduction of the EMG artifact. © 2000 IPPEM. Published by Elsevier Science Ltd. All rights reserved.

**Keywords:** Electrocardiogram; Electromyogram artifact; Filtering; Approximation filtering; Dynamic filtering

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**1. Introduction**

Various artifacts often contaminate electrocardiogram (ECG) recording, of which the most common are power-line interference and baseline drift. They were subjected to extensive research and many successful solutions have been found and applied.

The electromyogram (EMG) artifacts, as obtained from the same electrodes as the ECG, are difficult to remove, due to considerable overlapping of the frequency spectra of these two types of signals. EMG artifacts in ECG are quite common in subjects with uncontrollable tremor, in disabled persons having to exert effort in maintaining a position of their extremities or a body posture, in children, etc.

Attempts at filtering out the EMG were only partially successful. Therefore, a compromise was universally accepted and postulated in standards and recommendations, namely the application of a low pass filter with a cut-off frequency of minimum 35 Hz [1]. The result is a limited suppression of the EMG artifact and a reduction in the amplitudes of sharp ECG waves, such as

Q, R and S. This fact precludes accurate ECG diagnostic measurements if the EMG filter was applied [2].

Adaptive filtering of the EMG artifact was attempted with limited success. When the QRS complex disturbs the adaptation process, readaptation occurs and the artifact appears [3]. A modification of the algorithm could deal with this inconvenient, but the price again is a reduction of the sharp ECG amplitudes [4]. Better results with a faster convergence were obtained by Luo and Tompkins [5], but at the price of using an additional EMG channel.

Rossi et al. [6] proposed a low pass comb filter with reduced lobes in the frequency band above 50 Hz by cascading three averaging filters. In order to improve the reduced frequency response, they added a compensating filter gradually enhancing components from 10 to 100 Hz and having a first zero at 200 Hz. The goal was to obtain the  $-3$  dB cutoff at 35 Hz, as required by the standard. Unfortunately, no data on the compensating filter were given and no results with real signals were presented.

Levkov [7] also tried to use comb filters for simultaneous suppression of 50 Hz interference and EMG artifacts. However, reaching  $-3$  dB at 35 Hz resulted in an enhanced lobe with a maximum of 95% at 70 Hz.

In an attempt to improve the traditional compromise

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between efficient EMG artifact suppression and preservation of the ECG waveform, we applied approximation filtering, adding dynamic modification of the approximation function parameters depending on the ECG signal slew rate.

## 2. Method

The proposed method is based on the application of an approximation filter, whose characteristics are dynamically changed depending on the ECG signal slope. The implementation of this procedure requires an adequate evaluation of the slope regardless of the EMG noise component. For this purpose a function we called “wings” is applied. Then an appropriate relation is established between slope values and number of samples over which the approximation is to be accomplished.

The raw ECG signal is first subjected to an efficient 50 Hz subtraction procedure we have used for years [8,9]. Drift suppression can also be applied, but it is not necessary for the application of the proposed method. This is due to the fact that low frequency components of baseline fluctuations rarely exceed 0.5 Hz. However, when very strong drift is present, suppression procedures using for example backward filtering can be applied [9].

### 2.1. Signal slope evaluation

The “wings” function is found by multiplying the slopes of two adjacent segments of 10 ms length having a common point. Then the product absolute value is taken and inverted. For a 400 Hz sampling rate the “wings” function is:

$$\text{Wings} = W = -\text{abs}((S_i - S_{i-4})(S_i - S_{i+4})).$$

The segment length of 10 ms was chosen in order to obtain a certain level of signal smoothing in the process of slope evaluation. This value is not especially critical, as smaller interval results in more varying “wings” function and thus stronger additional smoothing is needed, and vice versa. Sliding this function along the signal, it reflects the variations of the angle between the two segments.

Well-shaped negative peaks of  $W$  correspond to every Q, R and S wave of the ECG. Negative peaks appear also at the maximum slew rate of QR and RS segments. An example is given in Fig. 1, where (a) is an artificial signal consisting of two waves of different slopes. The “wings” function is presented in (b) together with its smoothed version. The smoothing involves a running approximation procedure (discussed in section B), applied on segments of 35 ms length and a comb filtering with a first zero at 50 Hz and  $-3$  dB at 24 Hz. The result is a negative wave that reacts to the high slew rate

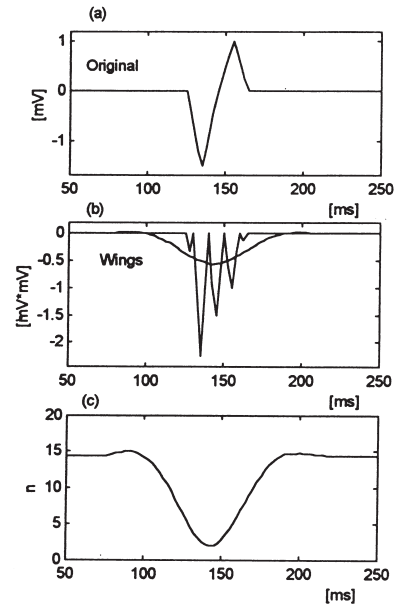


Fig. 1. (a) A synthesised QRS-like signal of two waves with different slopes and polarity. (b) The slope detection (“wings”) function and its smoothed version. (c) The time-course of the approximation interval  $n$  at both sides of a given sample (from  $n_{\min}=1$  to  $n_{\max}=15$ ), corresponding to the smoothed “wings” function.

elements (QRS complexes and/or T waves) of the ECG [Fig. 1(b)].

The slope evaluation of real ECG signals contaminated with EMG is preceded again by 50 Hz comb filtering and by a running approximation smoothing over 75 ms segments. Then the above-described procedure is applied.

### 2.2. Approximation

A smoothing procedure adopted from the work of Savitzky and Golay [10] is applied. It makes use of the least squares approximation method, applied for defining the weighting coefficients. The mathematical description of the process is:

$$Y_i = \frac{1}{N} \sum_{j=-n}^{j=n} C_j X_{i+j}$$

$Y$  and  $X$  represent the signal after and before approximation respectively.  $n$  is the length of the approximation interval at both sides of a sample.  $C_j$  are weighted approximation coefficients, and  $N$  is a normalization coefficient. The procedure is applied on  $2n+1$  samples.

The approximation coefficients are

$$C_j = 3n^2 + 3n - 1 - 5j^2$$

and the normalization coefficient is

$$N = \frac{(2n+1)(4n^2+4n-3)}{3}.$$

### 2.3. Dynamic approximation

Increasing the length of the approximation segment results in better EMG suppression, but greater amplitude reduction of the Q, R and S peaks will occur. This incited us to use approximation with adaptively adjusted segment length. Thus weaker smoothing is done on QRS complexes and/or T waves of high slew rate and greater on the remaining signal components. The values of the smoothed “wings” function are made to correspond to the approximation interval  $n$ . The time-course of  $n$  is given in Fig. 1(c).

If  $W$  is the smoothed “wings” and  $W_{\max}$ ,  $W_{\min}$  its maximum and minimum values, the transfer formula to  $n$ , varying from  $n_{\min}$  to  $n_{\max}$ , is:

$$n = n_{\min} + (n_{\max} - n_{\min}) \frac{W - W_{\min}}{W_{\max} - W_{\min}}.$$

The values  $n_{\min}=1$  to  $n_{\max}=15$  were chosen after careful testing of a multitude of ECG signals from our 400 records of 12-channel ECG databank.

### 2.4. EMG artifact

EMG signals were obtained from two ECG electrodes placed on one forearm. The ECG amplifier was used and the recordings were made during sustained voluntary effort. The artifacts thus obtained were weighted and additively mixed with the different ECG signals subjected to processing.

In order to have a basis for comparison while taking into account the requirement of the existing standard, we applied filtering with a 1st order of 35 Hz cutoff frequency filter. Adaptive filtering was also considered for comparison, using results of other authors.

The filtering efficiency was estimated by measuring the reduction of sharp Q, R or S amplitudes and the EMG suppression obtained. Due to the greatly varying EMG amplitude, the EMG filtering estimation was based on measurement of relatively well expressed EMG waves, for example the one preceding the QRS complex in Figs. 2(c) and 4. The ratio of original to suppressed EMG was adopted as a measure.

## 3. Results

The proposed method was tested on 400 records of a database collected with 400 Hz sampling rate and 4.88  $\mu\text{V/bit}$  resolution. As a difficult filtering example we selected a pure ECG signal (V1 lead) having sharpest QRS complexes of varying amplitude [Fig. 2(a), 1st trace]. It was mixed with EMG artifact of amplitude about 35% of the QRS [Fig. 2(b), 1st trace and Fig. 2(c)]. The corresponding approximation interval, scaled from

$n=1-15$ , is shown in Fig. 2(d). The result of the dynamic approximation is presented in Fig. 2(e). The R and S amplitudes in the second complex appear reduced by 0.06 mV, but the superimposed artifact cannot be accounted for. The EMG artifact depression ratio is 6.66.

Another example is given in Fig. 3, selected to show the performance of the method with shape-varying ventricular complexes. The same arrangement is used as in Fig. 2.

The application of a 35 Hz low pass 1st order Butterworth filter on the same ECG as in Fig. 2 results in a loss of 0.225 mV of the S-wave amplitude [Fig. 4(a)]. The EMG artifact suppression ratio is 2. The use of higher order filtering is presented in Fig. 4(b) and (c). Small  $r'$  waves after the QRS complex can be observed in Fig. 4(c). The EMG suppression ratio is 3.

Bidirectional filtering was also tested, in an attempt to avoid the ringing effect. However, with a 1st order filter, wave peaks and artifact reduction were virtually equal to those with a 2nd order mono-directional filter. We also tried adaptive filtering, but the results were not better. A comparison with results of other authors using adaptive procedures is discussed in the next section.

## 4. Discussion

In the preliminary stages of this study we applied various known filtering procedures: digitally implemented RC filters of different order, FFT based filtering, etc. Butterworth higher order filters showed better efficiency for EMG artifact suppression, therefore we used them for comparison.

We applied processing of synthesized triangular signals, as the one shown in Fig. 1(a) and other types previously described [2] and “clean” ECG signals, in order to evaluate the induced distortions. Mixing them with real EMG artifacts allowed for assessment of the suppression efficiency. This approach was applied on the 400 electrocardiograms from our 12-channel ECG database records.

The application of approximation filtering did not result in measurable peak amplitude losses of sharp Q, R or S waves in the majority of the 400 ECG records. In the example shown in Fig. 2 no R and S peak amplitude losses [Fig. 2(a) compared with Fig. 2(e)] could be detected in the first QRS complex. The second complex R and S waves were reduced by 0.06 mV. Obviously these differences contain a measurement error due to residual EMG superimposed on the processed signal. The distortions involved by the procedure can be seen in Fig. 2(a) (second trace), which represents the difference between the pure ECG (first trace) and its processed version. The difference amplitudes appear higher than the measured R and S reductions, due to the fact that they occur within the slopes, which are of lesser diagnostic

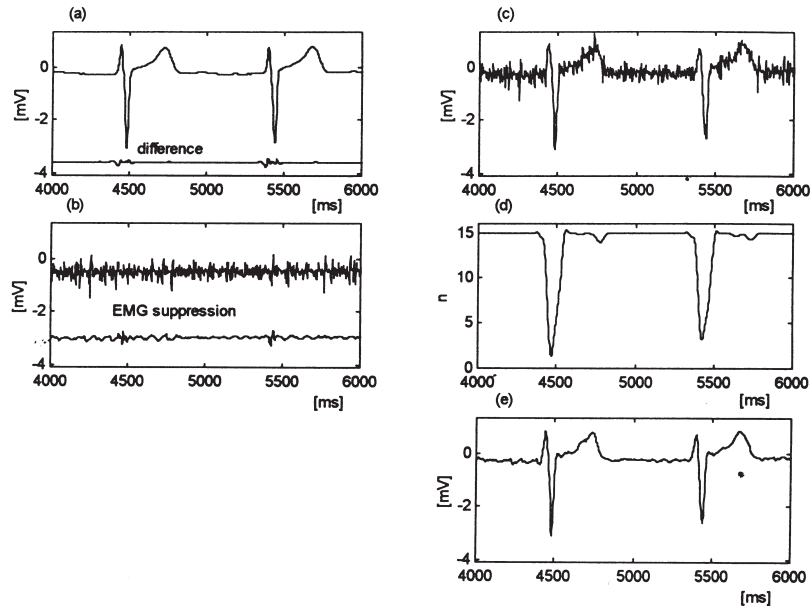


Fig. 2. Dynamic approximation procedure applied on a specially selected (out of a 400 records database) V1 lead ECG with sharp R and S waves. (a) A “pure” ECG signal (1st trace) and the difference with its processed version (second trace). (b) EMG artifact of amplitude about 35% of the QRS (1st trace) and the suppressed EMG after dynamic approximation (second trace). (c) The mixed ECG and EMG signal. (d) The corresponding approximation interval, scaled from  $n_{\min}=1$  to  $n_{\max}=15$ . (e) Result of the dynamic approximation applied on (c). The peak amplitudes of the steep R and S waves in the first complex are unchanged and reduced by 0.01 and 0.02 mV respectively and the EMG artifact is reduced several times.

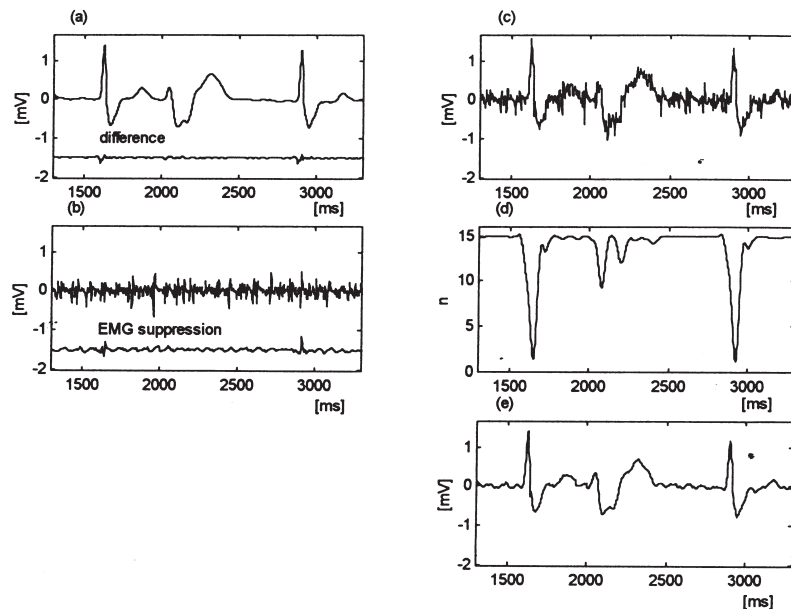


Fig. 3. Another example of the dynamic approximation procedure. (a) A “pure” ECG (first trace) and the difference with its processed version (second trace). (b) EMG interference (first trace) and the suppressed EMG after dynamic approximation (second trace). (c) The mixed ECG and EMG signal. (d) The corresponding approximation interval, scaled from  $n_{\min}=1$  to  $n_{\max}=15$ . (e) Result of the dynamic approximation. The peak amplitudes of the steep R and S waves of the first complex are preserved reduced by 0.128 and 0.064 mV respectively, and the EMG artifact is reduced several times.

significance. The EMG artifact suppression ratio is 6.66. It is higher outside the QRS complex interval and lower inside it, which is to be expected, according to the proposed method.

The example given in Fig. 3 was selected to show the performance of the method with shape-varying ventricular complexes and with a sharp R wave (40 ms duration, lead V5). The same arrangement is used as in Fig. 2. The

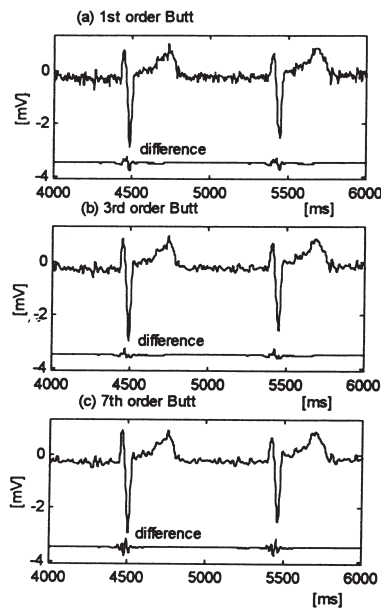


Fig. 4. The application of a 35 Hz low pass Butterworth filter on the same ECG as in Fig. 2. (a) 1st order Butterworth filtering results in 0.48 mV (23%) reduction of the S-wave amplitude. The EMG artifact reduction is about 2.5 times (first trace). The difference between the “pure” of Fig. 2(d) and its processed version is given in the second trace. (b) 3rd order Butterworth filter result presented as in (a). (c) 7th order Butterworth filter result presented as in (a).

processed signal R and S amplitudes showed elevated amplitude of about 0.07 mV in the first complex and 0.06 mV loss in the second. This result has an error due again to the artifact residue. The EMG depression retained its value of about 6.

The application of a 35 Hz low pass 1st order Butterworth filter on the same ECG as in Fig. 2 resulted in a reduction of 0.26 mV of the S-wave amplitude [Fig. 4(a)], while the EMG artifact suppression ratio was only 2, compared to  $>6$  with the dynamic approximation. The use of higher order filtering [Fig. 4(b) and (c)] yielded better results regarding reduction R and S amplitudes, but the well-known “ringing” effect was present. Small  $r'$  waves after the QRS complex can be observed in Fig. 4(c). The EMG suppression ratio is about 3, being twice lower than with the proposed procedure.

A more detailed observation of the difference signals [Figs. 2(a) and 3(a)—second traces] reveals that a slight distortion is induced in the first ascending component of the sharpest ECG waves. This leads to a 6 ms widening of the R-wave at the baseline, in the cases of Figs. 2(e) and 3(e). We observed widening and smoothing of some very small Q or  $r$ -waves of less than 0.1 mV amplitude. Having in mind the diagnostic criteria for ECG wave amplitudes and durations, these errors can be considered acceptable, compared to the obvious QRS amplitude reduction obtained by the currently applied 1st order filtering. On the other hand, it should be noted that the ECGs shown here were selected from the database as

ones with the sharpest R and S waves, being most difficult for EMG filtering. Obviously, complexes of lower slew rate had lesser or virtually no duration or amplitude changes. Amplitude changes in the range of 0.01–0.02 mV were found in most cases.

Thakor and Zhu [3] showed the application of adaptive filtering for reduction of EMG artifacts. As it can be seen in Fig. 5 of their work, after the QRS a readaptation process was started with the EMG artifact reaching 20% to 33% of the QRS amplitude and persisting till the T-wave. Better convergence was obtained by Bensadoun et al. [4], but at the price of twofold reduction of the Q-wave amplitude. Luo and Tompkins [5] showed very good artifact reduction, apparently with preservation of the ECG amplitudes. However, these results were obtained using a separate channel for feeding the artifact signal to the adaptive filter. This would be an important drawback in multichannel ECG recording.

## 5. Conclusion

The use of approximation filtering with dynamically varied number of samples and weighting coefficients, depending on the ECG signal slope, proved a relatively simple method. It yielded better waveform preservation and EMG artifact suppression, compared to most often used filtering approaches. The slope evaluation was implemented by sliding two adjacent segments along the signal and using the absolute value of the product of their slopes as a measure.

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